

One Last Theorem in the Family

We've built a chain of theorems throughout this course, each relating a **boundary integral** to an **interior integral**: Today: the **Divergence Theorem**, connecting a surface integral (flux *through* a closed surface) to a triple integral (divergence over the enclosed solid). It is the 3D analogue of Green's Theorem in *flux form*.

Every major theorem in Chapter 16 has the same shape:

$$\int_{\text{boundary}} (\text{something}) = \int_{\text{interior}} (\text{derivative of something}). \text{ The Divergence Theorem is the final version.}$$

The Divergence Theorem

Theorem: The Divergence Theorem (Gauss's Theorem)

Let E be a simple solid region, and let S be its boundary surface, oriented outward. Let $\vec{F}$ be a vector field whose components have continuous partial derivatives on an open region containing E . Then

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \operatorname{div} \vec{F} \, dV.$$

In components, with $\vec{F} = \langle P, Q, R \rangle$:

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dV.$$

Reading the two sides:

- **Left:** total flux of $\vec{F}$ outward through the closed surface S .
- **Right:** total divergence of $\vec{F}$ throughout the solid E .

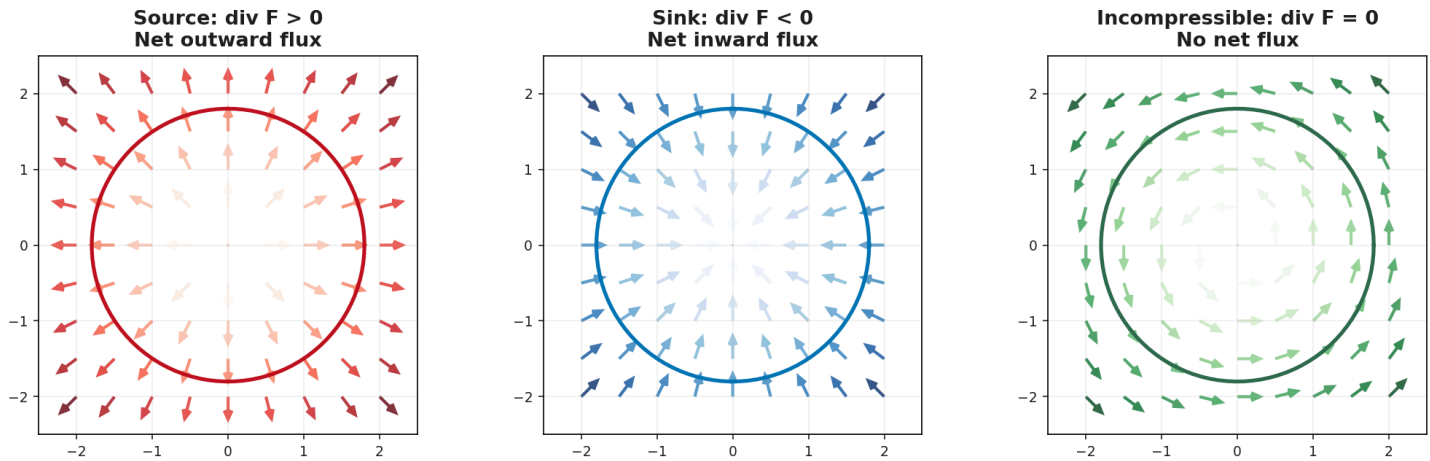
Flux *through the boundary* = total source strength *inside*. If $\operatorname{div} \vec{F} > 0$ inside E , net outward flux (source). If $\operatorname{div} \vec{F} < 0$, net inward flux (sink).

Physical Interpretation: Sources and Sinks

Think of $\vec{F}$ as the velocity field of a fluid. The Divergence Theorem says net outflow = total internal source strength:

$$\underbrace{\iint_S \vec{F} \cdot d\vec{S}}_{\text{net flow out through } S} = \underbrace{\iiint_E \operatorname{div} \vec{F} \, dV}_{\text{total source/sink inside } E}.$$

Physical Interpretation of Divergence



Divergence is the *local expansion rate* of the field. The theorem sums all those local expansions to produce the net boundary outflow.

Connection to Green's Theorem (Flux Form)

Recall the **flux form of Green's Theorem** (16.4):

$$\oint_C \vec{F} \cdot \vec{n} \, ds = \iint_D \text{div } \vec{F} \, dA \quad (\vec{F} = \langle P, Q \rangle, \text{ 2D}).$$

The Divergence Theorem is the **same statement, one dimension up**:

Green's (flux form) is the 2D Divergence Theorem. Every theorem in Chapter 16 follows one pattern: integral over boundary = integral of a derivative over interior.

Quick Check: Compute the Divergence

Compute $\text{div } \vec{F}$ for each field:

1. $\vec{F} = \langle x^2, xy, z \rangle$
2. $\vec{F} = \langle x^3 + y^3, y^3 + z^3, z^3 + x^3 \rangle$
3. $\vec{F} = \langle yz, xz, xy \rangle$

Recall: $\text{div } \vec{F} = P_x + Q_y + R_z$.

Example 1: Flux Through a Paraboloid Region

Example 1

Use the Divergence Theorem to find the flux of $\vec{F} = \langle x^2, xy, z \rangle$ outward through the closed surface S bounding the solid E below $z = 4 - x^2 - y^2$ and above $z = 0$.

Computing the flux directly is painful: the boundary S has two pieces (paraboloid cap + flat disk below), each needing its own parametrization. Use the Divergence Theorem to replace it with one triple integral over E . Cylindrical coordinates work here.

Quick Check: Which Side Is Easier?

For each scenario, decide which side of the Divergence Theorem is easier to compute.

1. $\vec{F} = \langle x, y, z \rangle$, S is a complicated surface enclosing a solid of known volume.
2. $\operatorname{div} \vec{F}$ is a messy expression, but S is a single simple surface.
3. $\vec{F} = \langle xz^2, x^2y, y^2z \rangle$, E is the unit cube $[0, 1]^3$.

Ask: is the divergence simple? Is the surface a single piece? Is the volume easy to compute?

Example 2: Flux Through a Sphere

Example 2

Use the Divergence Theorem to find the flux of $\vec{F} = \langle x^3 + y^3, y^3 + z^3, z^3 + x^3 \rangle$ outward through the sphere S of radius 2 centered at the origin.

The divergence is $3(x^2 + y^2 + z^2) = 3\rho^2$ in spherical coordinates — a gift. Set up the triple integral over the ball.

Example 3: Flux Through a Cube

Example 3

Find the flux of $\vec{F} = \langle xy, y^2, yz \rangle$ outward through the surface of the cube $[0, 1]^3$.

The cube has SIX faces. Direct computation means six separate surface integrals, each with its own outward normal. Use the Divergence Theorem.

Quick Check: The Radial Field Through a Sphere

Let $\vec{F} = \langle x, y, z \rangle$ and S be the unit sphere. Find $\iint_S \vec{F} \cdot d\vec{S}$ without parametrizing anything.

Compute $\text{div } \vec{F}$. If it's a constant c , the flux is just $c \cdot \text{Vol}(E)$.

When NOT to Use the Divergence Theorem

The theorem requires a **closed** surface enclosing a solid with $\vec{F}$ smooth throughout. Common pitfalls:

1. Open surface. E.g., a hemisphere with no disk on the bottom. *Fix:* add the missing piece, apply the theorem to the closed surface, subtract the flux through the added piece.

2. Singularity inside E . E.g., $\vec{F} = \vec{r}/|\vec{r}|^3$ with the origin inside. *Fix:* cut out a small sphere around the singularity and treat the region as topologically a shell.

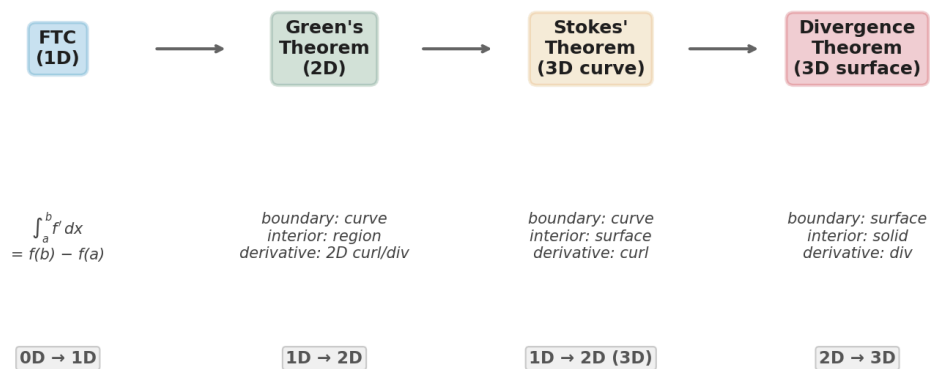
3. Non-simple regions (cavities or holes). *Fix:* decompose into simple pieces.

Checklist before applying the theorem: (i) Is S closed? (ii) Is $\vec{F}$ smooth throughout E ? (iii) Is S oriented *outward*?
Failure on any one means the theorem needs adaptation.

The Big Picture: Four Theorems, One Pattern

Every major theorem in vector calculus is a version of:

$$\int_{\text{boundary}} (\text{something}) = \int_{\text{interior}} (\text{derivative of something}).$$



Unifying Principle: integral over boundary = integral of derivative over interior

FTC, Green's, Stokes', Divergence are not four separate theorems — they are the *same theorem* in different dimensions.

Which Theorem? A Decision Tree

Decision tree for integrals in Chapter 16:

1. $\vec{F}$ conservative? $\Rightarrow$ **FTC for line integrals**: find f , evaluate at endpoints.
2. Line integral around a closed curve in the plane? $\Rightarrow$ **Green's**.
3. Line integral around a closed curve in 3D? $\Rightarrow$ **Stokes'**.
4. Flux through a closed surface in 3D? $\Rightarrow$ **Divergence**.

Example 4

Match each integral to the theorem that simplifies it:

- [(a)] $\iint_S \vec{F} \cdot d\vec{S}$, S the closed surface of a cylinder, with $\operatorname{div} \vec{F} = 5$.
- [(b)] $\oint_C \vec{F} \cdot d\vec{r}$, C the boundary of a triangle in space.
- [(c)] $\int_C \nabla f \cdot d\vec{r}$ from $(0, 0)$ to $(3, 4)$.

Map each integral to one of the four theorems.

Homework

Section 16.9: 1, 3, 5, 7, 11